

# Nikhil T

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## PROFESSIONAL SUMMARY

Results-driven Software Engineer with strong experience in full-stack development, AI/ML, and cloud-native systems, skilled in Python, Java, JavaScript, TypeScript, Spring Boot, React/Next.js, Node.js, and FastAPI. Proficient in building scalable microservices, REST/GraphQL APIs, distributed systems, event-driven architectures using Kafka and RabbitMQ, and deploying applications on AWS with Docker and Kubernetes. Experienced in developing RAG pipelines, Generative AI solutions, computer vision models, and data-driven analytics using TensorFlow, OpenCV, FAISS, BM25, and Pandas. Demonstrated ability to improve system performance, automate workflows, optimize data pipelines, and deliver high-quality software in Agile environments. Strong communicator with experience mentoring students, contributing to research, and leading end-to-end project delivery.

## SKILLS

**Programming Languages:** Python, Java, JavaScript, TypeScript, C++

**Web Development:** React.js, Next.js, Spring boot 3.x, Node.js, Express, FastAPI, TailwindCSS, HTML5, CSS3

**Databases:** MySQL, MongoDB, PostgreSQL, ChromaDB, PineCone, Redis

**Messaging & Streaming:** Apache Kafka, RabbitMQ

**System Design Concepts:** Microservices, Load Balancing, WebSockets, Caching, Distributed Systems

**DevOps/Cloud:** Git, GitHub, Docker, Kubernetes, AWS (EC2, Lambda, S3, RDS), Github Actions, Jenkins

**AI/ML:** Generative AI, LangChain, RAG, CNN, Computer Vision, NLP, LangGraph, AI Tooling (Cursor, Claude Code, MCP)

**Methodologies:** Agile, Scrum, CI/CD, REST/GraphQL API Design

## WORK EXPERIENCE

### UNIVERSITY OF NORTH TEXAS

Denton, Texas, USA

#### Teaching Assistant

Jan 2025 – Present

- Delivered **13-14 lab sessions per semester** across **multiple sections**, supporting **200+ undergraduate students** in Data Structures and Algorithms through hands-on coding exercises and concept reinforcement.
- Graded **40+ students per assignment and exam**, providing structured feedback that contributed to a **~15% improvement in average scores** and a **~12% increase in section pass rates** over the semester.
- Developed and maintained **course materials including 20+ coding exercises and practice problems**, enhancing student understanding of core programming concepts across multiple sections.
- Provided **one-on-one tutoring** and clarified algorithmic concepts during lab discussions, directly supporting students in mastering core programming fundamentals.

### ORIANA SOFTWARE SOLUTIONS

Hyderabad, India

#### Software Intern

June 2023 – July 2024

- Developed a **syndicated loans platform** using **Spring Boot, React.js, and MongoDB**, improving transaction efficiency and workflow automation by **~45-50%**.
- Implemented a **Kafka-based event-driven architecture**, reducing system latency and ensuring seamless communication between distributed microservices.
- Collaborated with a **5-member Agile team** to build secure REST APIs, implement authentication modules, and integrate third-party financial data services.
- Enhanced **UI/UX and data synchronization pipelines**, increasing user adoption and engagement by **~30%** during pilot rollout.
- Contributed to the **first-of-its-kind fintech product** in the company's portfolio, enabling streamlined multi-party loan participation and supporting a projected **25% market expansion**.

## RESEARCH PAPER

- Sahith.T, Nikhil.T, Sai Rashmitha.S, Sandhya.S “*Real-Time Multiple Face Detection using Multi-Task Convolutional Neural Networks*”, Industrial Engineering Journal, ISSN: 0970-2555 Volume: 52, Issue 8, August: 2023, UGC CARE Group-1.
- Link: [http://www.journal-iiie-india.com/1\\_aug\\_23/33\\_online.pdf](http://www.journal-iiie-india.com/1_aug_23/33_online.pdf)

## PROJECT EXPERIENCE

### LangGraph Stock Pattern Analyzer (FinBot) | Python, LangGraph, Google Gemini API, yfinance, Pandas, Matplotlib

- Built an **AI-powered stock analysis platform** using **LangGraph and Gemini API**, enabling real-time pattern recognition and reasoning-based market summaries.
- Processed **700+ live OHLCV datapoints per symbol** and detected complex chart formations with **92% pattern accuracy** (Double/Triple Tops, Head & Shoulders, Wedges, Triangles, etc.).
- Reduced manual analysis time by **80%** through LLM-powered insights with full **memory traceability** and **prompt logging** for transparency.
- Designed modular, state-driven architecture using **LangGraph's StateGraph** and **TypedDict**, enhancing scalability and fault tolerance.

### Customer Support RAG System | Python, FAISS, Sentence Transformers, BM25, Cross Encoder, Gradio, OpenAI

- Developed a **hybrid RAG pipeline** that improved response accuracy by **35%** through **dense (FAISS)** and **sparse (BM25)** retrieval with reranking via **Cross Encoder (MiniLM-L-6-v2)**.
- Indexed and embedded **10,000+ document chunks** for contextual Q&A, cutting average response time by **60%**.
- Designed **prompt-engineered guardrails** to reduce hallucinations by **70%**, ensuring fact-based customer support automation.
- Delivered an intuitive **Gradio UI** with ranked results, retrieval trace visualization, and JSON-based audit logging.

### Welding Pro – Microservices-Based E-Commerce Platform | Java, Spring Boot, Next.js, Docker, Netlify, WebSockets

- Designed and deployed a **scalable microservices e-commerce system** for welding products and services, handling **100,000+ transactions** during Phase 1.
- Implemented **secure user authentication, payment gateway integration, product management, and real-time order tracking** via WebSockets.
- Led a **team of four developers**, ensuring Agile sprint delivery and seamless integration of microservices.

- Deployed using **Next.js, Docker and Netlify**, achieving 99.8% uptime and onboarding 10+ business users in the pilot rollout.

#### NYC Pedestrian & Subway Traffic Analysis | Python, Tableau, Power BI, Pandas, NumPy

- Developed an interactive **data visualization and analytics platform** examining **15+ years of NYC pedestrian and subway infrastructure data (2007–2022)** using Tableau and Power BI.
- Processed and geo-encoded **1.2M+ pedestrian and subway records** with Python (Pandas, NumPy), optimizing data pipelines and improving visualization accuracy by **30%**.
- Conducted **temporal, spatial, and trend analyses** to reveal key insights — including **2.5× higher evening foot traffic** and **~60% pedestrian concentration in Manhattan** — supporting **urban mobility planning** and infrastructure optimization.
- Built **multi-dashboard analytics with interactive heatmaps, time-series visualizations, and geographic insights**, enabling decision-makers to identify **high-traffic zones** and **seasonal flow variations** in real time.

#### Cuisine Connect – Restaurant Management Portal | MERN Stack, Kafka, Node.js, MongoDB, React.js

- Spearheaded the development of **Cuisine Connect**, a sophisticated **MERN stack-based restaurant management system**.
- Optimized order processing and real-time operational tracking, improving **customer satisfaction by 30% and operational efficiency by 25%**.
- Integrated **Kafka** for real-time updates and implemented advanced **scalability features** supporting **500+ daily transactions**.

#### ElderCare Voice Assistant | Python, Tkinter, Groq LLM API, SpeechRecognition

- Developed an **AI-powered health management system** for seniors, integrating **voice-enabled interaction** and an **intuitive GUI** to track glucose levels, sleep, and medications with minimal technical effort.
- Engineered **speech-to-text and text-to-speech pipelines** using **SpeechRecognition** and **Pytsx3**, achieving **>90% voice command accuracy** and real-time response through Groq's LLM API.
- Built **modular Python components for health data tracking, medication reminders, and emergency alerts**, automating recurring tasks via **Schedule** and ensuring robust logging for system monitoring.
- Designed an **accessibility-focused interface** with Tkinter, featuring **high-contrast modes, adaptive fonts, and quick emergency actions**, enhancing usability for **elderly users aged 60+**.

#### Cold Email Generator | Python, LangChain, Groq (LLaMA 3), ChromaDB, Streamlit

- Engineered an **AI-driven automation system** that **scrapes and analyzes 500+ job postings** from company career pages, generating personalized cold emails tailored to each role.
- Integrated **LLaMA 3 via Groq API** for **LLM-based job description parsing and email synthesis**, achieving **85% accuracy in role-content relevance** and reducing manual drafting time by **90%**.
- Designed and deployed a **Streamlit web interface** enabling instant, one-click email generation with dynamic previews and role-based tone customization, improving **outreach productivity by 4×**.
- Implemented **semantic portfolio matching** through **ChromaDB vector search** over 200+ portfolio items, ensuring high contextual alignment between user projects and target job roles.
- Developed **modular LangChain pipelines** for scraping, LLM prompt chaining, and vector retrieval; optimized execution latency to under **3 seconds per request** with persistent local vector storage.

#### Market Regime Detection System | Python, Yahoo Finance API, AWS S3, Streamlit, Plotly, Pandas

- Developed **modular LangChain pipelines** for scraping, LLM prompt chaining, and vector retrieval; optimized execution latency to under **3 seconds per request** with persistent local vector storage.
- Engineered a **real-time financial analytics system** analyzing **12+ years of market data** (SPY, VIX, TIP, IEF, SHY) to classify regimes such as *Growth*, *Risk-Off*, and *Inflation*.
- Automated **data ingestion, computation, and visualization pipelines** with Pandas, NumPy, and Plotly, achieving **95% data accuracy** and **sub-5-second latency** per refresh.
- Developed an **interactive Streamlit dashboard** integrated with **AWS S3** for live tracking of macro indicators and sector rotations across **8+ market sectors**.
- Implemented a **confidence-scored regime model** and sector rotation logic yielding **85% alignment with institutional macro trends**, improving decision-support accuracy by **40%**.

#### Multi-Face Detection and Recognition System | Python, MTCNN, ResNet, OpenCV, TensorFlow

- Engineered a **multi-face recognition pipeline** using **MTCNN and ResNet**, achieving **98% detection accuracy** and **15% reduction in false positives** over baseline models.
- Implemented optimized **face alignment and feature extraction** workflows, improving inference speed by **25%**.
- Applied **data augmentation and fine-tuning techniques**, ensuring robustness under varying lighting and occlusion conditions.
- Enhanced security and performance benchmarking across **1,000+ test images**, outperforming standard face recognition benchmarks.

#### form-sanitize | JavaScript, TypeScript, Node.js, React, Express | [LINK](#)

- Built and published a **schema-based form sanitization library** on **npm** achieving **100+ monthly downloads**, enabling shared validation logic across **React frontends and Express backends** with a single rule definition — eliminating duplicated sanitization code.
- Designed a **chainable, fluent API** with **10+ sanitization methods** including `.trim()`, `.stripTags()`, `.normalizeEmail()`, `.clamp()`, and `.toSlug()` across string, number, and boolean types, achieving **zero external dependencies**.
- Implemented **recursive nested object sanitization** via `s.object(definition)`, enabling deep sanitization of complex form payloads in a single `.sanitize()` call.
- Engineered **universal runtime compatibility** - works in both **Node.js and browser environments** — outperforming alternatives like DOMPurify and validator.js in schema support and portability.

#### DebtLens | TypeScript, Groq, Cloudflare Workers | [LINK](#)

- Developed and shipped a **VS Code extension** that transforms invisible tech debt into a **color-coded file tree heatmap** across **JavaScript/TypeScript** codebases, combining **Git age, TODO/FIXME density, and deprecated API detection** into a single **0–100 weighted debt score** across **5 severity levels**.
- Delivered **LLM-powered, real-time refactoring plans** via **Groq** with **10 free AI explanations/day** and zero configuration, lowering the barrier for teams to adopt **AI-assisted code quality workflows**.
- Engineered a **secure Cloudflare Worker proxy** ensuring no user code or API keys are ever exposed, with **bring-your-own-key** support for unlimited usage.

Orvexa | Next.js, FastAPI, Socket.io, Claude, Redis

- Built a **real-time collaborative AI research platform** where multiple users join shared rooms, chat live and trigger an AI agent via @ARIA that searches the web, summarizes findings, and streams results onto a **shared visual board** visible to all participants simultaneously.
- Engineered a **3-agent parallel pipeline** - search (Tavily), summarizer, and fact-checker - powered by **Claude Sonnet**, executing concurrently with **30s per-agent timeouts** and streaming responses word-by-word onto the board via **Redis pub/sub**.
- Architected a **real-time event layer** using **Socket.io** with room-based presence, typing indicators, and a structured **15+ socket event system** keeping all participants in sync across the shared board and chat.
- Designed a **fault-tolerant streaming system** with per-card 60s safety timers, interruption resilience, and automatic error cards - ensuring partial results always surface even when individual agents fail.

AI Creator Studio | Python, Streamlit, OpenAI (GPT-4o-mini), FAISS, Sentence Transformers, Supabase, bcrypt | [Live Demo](#)

- Built 4-station interactive web app teaching high school students how to construct a mini AI from scratch, covering system prompts, RAG pipelines, hallucination grounding, multi-turn chat memory, and prompt injection defense through hands-on exercises
- Engineered a hybrid RAG pipeline using FAISS (IndexFlatIP) and all-MiniLM-L6-v2 embeddings with 800-char chunking and 100-char overlap, delivering side-by-side grounded vs. ungrounded comparisons with cosine similarity scoring and retrieval trace visualization.
- Developed a Hacker Mode prompt injection playground with 4 attack techniques (classic override, role-play escape, context stuffing, jailbreak framing) and an LLM-as-judge evaluation loop using a second GPT-4o-mini call, returning breach/defend verdicts with a full hardening and defense re-run phase.
- Architected a Teacher Dashboard with bcrypt-hashed auth, Supabase (PostgreSQL) backend, class code generation, student enrollment flow, and real-time per-student station progress tracking across concurrent classrooms.
- Shipped 127 passing tests across 10 test files covering auth, RAG pipeline, LLM client, Supabase CRUD, classroom services, and full end-to-end integration — all with zero live API or network calls.

EDUCATION

|                                                                                        |            |
|----------------------------------------------------------------------------------------|------------|
| University of North Texas, Denton, Texas, USA                                          | 2026       |
| Master of Science - Computer Science                                                   |            |
| Relevant Coursework: Software Engineering, Big Data and Data Science, Database Systems | CGPA: 3.90 |
| VNR Vignana Jyoti Institute of Engineering & Technology, Hyderabad, India              | 2024       |
| Bachelor of Technology - Computer Science & Engineering                                | CGPA: 3.44 |

ACHIEVEMENTS

- **Winner, CFG Competition** — VNR College of Engineering
- **Runner-Up, Hackathon** — Vasavi College of Engineering

CERTIFICATIONS

- NPTEL Certification in **Big Data Computing** with **Elite Grade** and received **silver medal** from **IIT Kanpur**, 2022.
- Coursera Certification in **IT Automation with Python** from **Google**.